



Solution Set

Question 1

Then

$$z_1 = a + ib, \quad z_2 = c + id$$

$$\Re(z_1 z_2) = ac - bd$$

Question 2

$$z_1 + z_2 + 1 = 4, \quad z_1 - z_2 + 1 = 2 - 2i$$
$$\left| \frac{4}{2-2i} \right| = \sqrt{2}$$

Question 3

so the real part is

$$\frac{z_1 z_2}{z_1} = \frac{-3+4i}{2+i} = \frac{-2+11i}{5}$$

and

$$-\frac{2}{5}$$

$$\Im\left(\frac{1}{z_1 z_1}\right) = 0$$

Question 4

Hence the modulus of the difference is

$$\frac{1+i}{1-i} = i, \quad \frac{1-i}{1+i} = -i$$

$$|2i| = 2$$

Question 5

If

$$|\alpha| = |\beta| = 1$$

then

$$|1 - \bar{\alpha}\beta| = |\beta - \alpha|$$

so the required modulus is 1.

Question 6

$$|1 - i|^x = 2^x \Rightarrow 2^{x/2} = 2^x \Rightarrow x = 0$$

There is no non-zero integer solution.

Question 7

Real coefficients force conjugate roots. The roots are

$$1 + i, 1 - i, 2 - i, 2 + i$$

and the sum is

$$6$$

Question 8

The cube roots of unity are

$$1, \quad \omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad \omega^2 = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$$

and

$$1 + \omega + \omega^2 = 0$$